

```
!pip install transformers torch pandas matplotlib
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.10.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.17.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.29.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.4.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.7.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2025.11.12)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.10.1)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
```

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.20.0)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

print("Model Loaded Successfully!")
```

[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your notebook.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% 629/629 [00:00<00:00, 5.68kB/s]
model.safetensors: 100% 268M/268M [00:02<00:00, 130MB/s]
Loading weights: 100% 104/104 [00:00<00:00, 3343.71it/s]
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 4.44kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 13.9MB/s]
Model Loaded Successfully!
```

```
review = "This product is amazing and works perfectly."

result = classifier(review)

print(result)
```

```
[{'label': 'POSITIVE', 'score': 0.999881386756897}]
```

```
review = input("Enter a review: ")

result = classifier(review)
```

```
print("\nSentiment:", result[0]['label'])  
print("Confidence Score:", round(result[0]['score'], 4))
```

Enter a review: I love this product

Sentiment: POSITIVE
Confidence Score: 0.9999

```
import pandas as pd  
  
reviews = [  
    "This product is amazing.",  
    "I hate this service.",  
    "The experience was average.",  
    "Very good quality.",  
    "Worst purchase ever."  
]  
  
results = []  
  
for review in reviews:  
    sentiment = classifier(review)[0]  
  
    results.append([  
        review,  
        sentiment['label'],  
        round(sentiment['score'], 4)  
    ])  
  
df = pd.DataFrame(  
    results,  
    columns=["Review", "Sentiment", "Confidence"]  
)  
  
df
```

	Review	Sentiment	Confidence	
0	This product is amazing.	POSITIVE	0.9999	
1	I hate this service.	NEGATIVE	0.9997	
2	The experience was average.	NEGATIVE	0.9260	
3	Very good quality.	POSITIVE	0.9999	
4	Worst purchase ever.	NEGATIVE	0.9998	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
import matplotlib.pyplot as plt

sentiment_counts = df["Sentiment"].value_counts()

plt.figure(figsize=(5,5))
plt.pie(
    sentiment_counts,
    labels=sentiment_counts.index,
    autopct="%1.1f%%"
)

plt.title("Sentiment Distribution")
plt.show()
```

Sentiment Distribution

NEGATIVE



```
# Save results to CSV
```

```
df.to_csv("sentiment_results.csv", index=False)
```

```
print("Results saved successfully!")
```

Results saved successfully!

```
from google.colab import files
```

```
files.download("sentiment_results.csv")
```

POSITIVE

